



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
*SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN***

TECHNICAL MATHEMATICS P2/TEGNIESE WISKUNDE V2

MAY/JUNE 2026/MEI/JUNIE 2026

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

CODE/ KODE	EXPLANATION/VERDUIDELIKING
A	Accuracy/Akkuraatheid
AO	Answer only/Slegs antwoord
CA	Consistent accuracy/Volgehoue akkuraatheid
I	Identity/Identiteit
M	Method/Metode
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for omitting units/Geen penalisering vir eenhede weggelaat nie
R	Rounding/Afronding
RE	Reason/Rede
S	Simplification/Vereenvoudiging
SF	Substitution in correct formula/Vervanging in korrekte formule
ST	Statement / Bewering
ST/RE	Statement with reason/Bewering met rede
F	Correct formula/Korrekte formule

**These marking guidelines consist of 30 pages.
*Hierdie nasienriglyne bestaan uit 30 bladsye.***

NOTE:

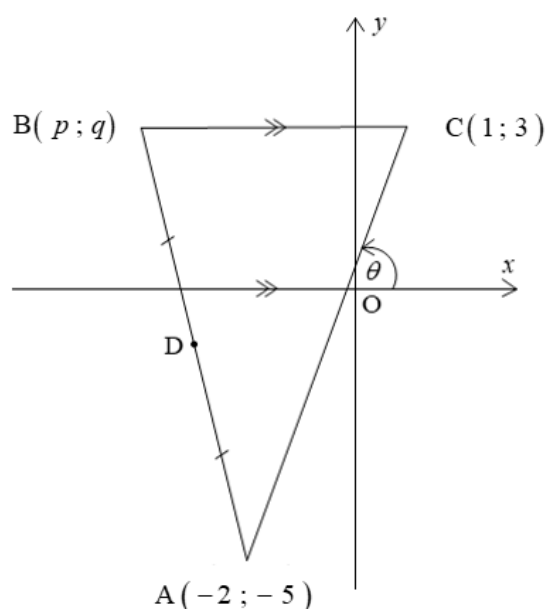
- If a candidate answers a question **TWICE**, only mark the **FIRST** attempt.
- The method of Consistent Accuracy marking must be applied in all aspects of the marking guideline where indicated with the marking code **CA**.

LET WEL:

- Indien 'n kandidaat 'n vraag **TWEE** keer beantwoord, sien slegs die **EERSTE** poging na.
- Die metode van Volgehoue Akkuraatheid-nasien moet waar moontlik op alle aspekte van die nasienriglyne toegepas word soos aangedui deur die nasienkode **CA**.

QUESTION/VRAAG 1

1.1



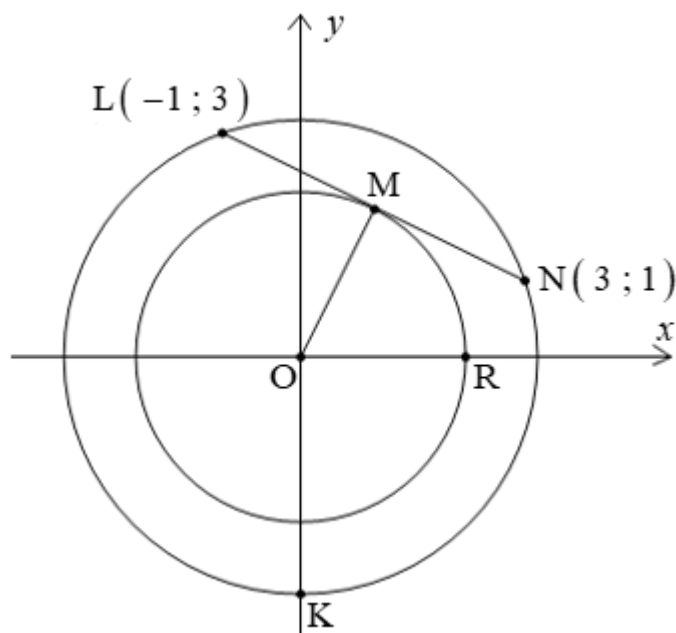
1.1.1	$q = 3$	✓ 3	A (1)
1.1.2	$p = -4$	✓ -4	A (1)
1.1.3	$m_{AC} = \frac{y_C - y_A}{x_C - x_A}$ $= \frac{3 - (-5)}{1 - (-2)}$ $= \frac{8}{3} \approx 2,67$	✓ SF ✓ gradient/gradiënt AO: full marks / volpunte	A CA (2)

1.1.4	$\tan \theta = m_{AC}$ $\theta = \tan^{-1}\left(\frac{8}{3}\right)$ OR / OF $\theta = \tan^{-1}(2,67)$ $\therefore \theta \approx 69,44^\circ$ OR / OF $\theta \approx 69,47^\circ$	✓ SF CA ✓ value of / waarde van θ CA AO: full marks / volpunte (2)
1.1.5	$D\left(\frac{x_A + x_B}{2}; \frac{y_A + y_B}{2}\right)$ $D\left(\frac{-2 + (-4)}{2}; \frac{-5 + 3}{2}\right)$ $D(-3; -1)$	✓ x-coordinate/koördinaat CA ✓ y-coordinate/koördinaat CA (2)
1.2.1	$2x - 5y = 10$ $y = \frac{2x - 10}{5}$ OR / OF $y = \frac{10 - 2x}{-5}$ OR / OF $\therefore y = \frac{2}{5}x - 2$ OR / OF $y = 0,4x - 2$	✓ make y the subject / maak y die onderwerp A OR / OF ✓ make y the subject / maak y die onderwerp A (1)
1.2.2	$\therefore m_{\perp} = -\frac{5}{2} = -2,5$	✓ m of perp. line / m van loodregte lyn CA (1)
1.2.3 #	$y - (0) = -\frac{5}{2}(x - (5))$ OR / OF $0 = -\frac{5}{2}(5) + c$ $c = \frac{25}{2}$ $\therefore y = -\frac{5}{2}x + \frac{25}{2}$ OR / OF $y = -2,5x + 12,5$	✓ SF CA ✓ equation / vergelyking CA (2)

[12]

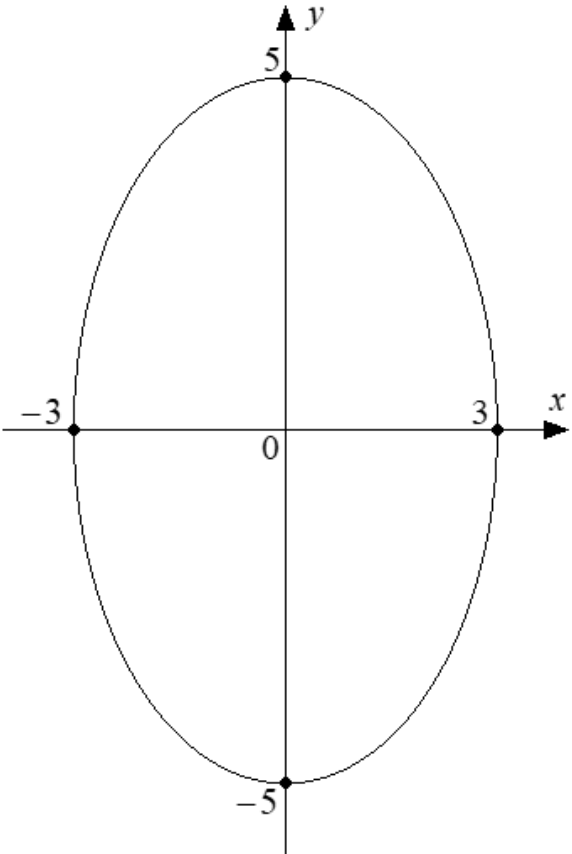
QUESTION/VRAAG 2

2.1



2.1.1	$x^2 + y^2 = r^2$ $(-1)^2 + (3)^2 = r^2$ OR / OF $(3)^2 + (1)^2 = r^2$ $r^2 = 10$ $\therefore x^2 + y^2 = 10$ OR / OF $\therefore x = \pm \sqrt{10 - y^2}$ OR / OF $\therefore y = \pm \sqrt{10 - x^2}$ OR / OF $x^2 + y^2 = (-1)^2 + (3)^2$ OR / OF $(3)^2 + (1)^2$ $= 10$	✓ SF L or / of N A ✓ equation/vergelyking CA OR / OF ✓ SF L or / of N A ✓ equation/vergelyking CA OR / OF ✓ SF L or / of N A ✓ equation/vergelyking CA OR / OF ✓ SF L or / of N A ✓ equation/vergelyking CA AO: full marks / volpunte (2)
2.1.2	K $(0; -\sqrt{10})$ OR / OF K $(0; -3,16)$	✓ $x_K = 0$ A ✓ $y_K = -\sqrt{10}$ CA (2)
2.1.3	Radius $\perp$ Tangent / Raaklyn	✓ RE A (1)

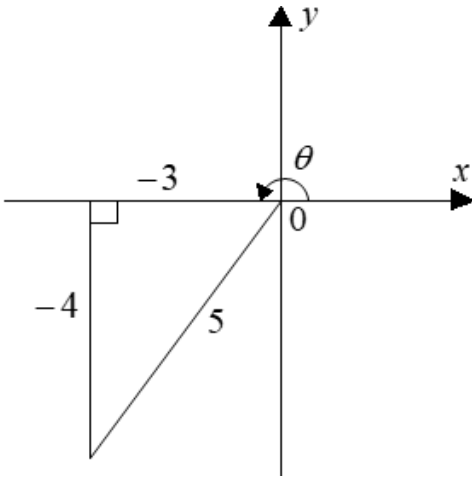
2.1.4	$\therefore M\left(\frac{-1+3}{2}; \frac{3+1}{2}\right)$ $M(1; 2)$ $\therefore$ Equation of smaller circle/<i>Vergelyking van kleiner sirkel</i>: $x^2 + y^2 = (1)^2 + (2)^2$ $x^2 + y^2 = 5$ $r = \sqrt{5}$ $\therefore R(\sqrt{5}; 0) \text{ OR / OF } R(2, 24; 0)$ OR / OF $\therefore M\left(\frac{-1+3}{2}; \frac{3+1}{2}\right)$ $M(1; 2)$ $OM = \sqrt{(0-1)^2 + (0-2)^2}$ $OM = \sqrt{5}$ $\therefore R(\sqrt{5}; 0) \text{ OR / OF } R(2, 24; 0)$	✓ SF A ✓ coordinates of/ <i>koördinate van M</i> CA ✓ SF CA ✓ $r^2 = 5$ OR / OF CA ✓ $r = \sqrt{5}$ CA ✓ coordinates of/ <i>koördinate van R</i> CA OR / OF ✓ SF A ✓ coordinates of/ <i>koördinate van M</i> CA ✓ SF CA ✓ length of/<i>lengte van</i> OM CA ✓ coordinates of/ <i>koördinate van R</i> CA AO: full marks / volpunte (5)
2.2.1	$\frac{x^2}{(3)^2} + \frac{y^2}{(5)^2} = 1$ OR / OF $\frac{x^2}{(\sqrt{9})^2} + \frac{y^2}{(\sqrt{25})^2} = 1$	✓ standard form / <i>standaard vorm</i> A OR / OF ✓ standard form / <i>standaard vorm</i> A (1)

2.2.2		✓ x- and y-intercepts / x- en y- afsnitte CA ✓ elliptical shape / <i>elliptiese vorm</i> CA (2)
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QUESTION/VRAAG 3

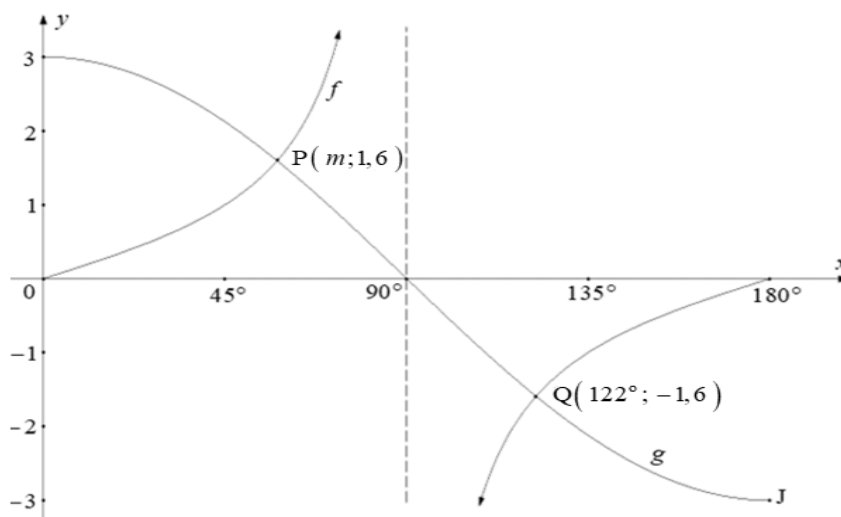
3.1.1	$T = 3,2\pi \times \frac{180^\circ}{\pi} = 576^\circ$	✓ T in degrees/grade A (1)
3.1.2	$\begin{aligned} &\text{cosec}(T - R) \\ &= \text{cosec}(576^\circ - 22,8^\circ) \\ &= \frac{1}{\sin 553,2^\circ} \\ &\approx -4,38 \end{aligned}$ OR / OF $R = 22,8^\circ \times \frac{\pi}{180^\circ} = \frac{19}{150}\pi \approx 0,40 \text{ rad}$ $\begin{aligned} &\text{cosec}(T - R) \\ &= \text{cosec}\left(3,2\pi - \frac{19}{150}\pi\right) \\ &= \frac{1}{\sin\left(\frac{461}{150}\pi\right)} \\ &\approx -4,38 \end{aligned}$	$\begin{aligned} &\checkmark \text{ substitution/substitusie} \quad \textbf{CA} \\ &\checkmark \textbf{I} \quad \textbf{A} \\ &\checkmark \text{ ratio/verhouding} \quad \textbf{CA} \end{aligned}$ OR / OF $\begin{aligned} &\checkmark \text{ substitution/substitusie} \quad \textbf{CA} \\ &\checkmark \textbf{I} \quad \textbf{A} \\ &\checkmark \text{ ratio/verhouding} \quad \textbf{CA} \end{aligned}$ AO: full marks / volpunte (3)

3.2		
	$(-3)^2 + y^2 = (5)^2$ $y = -4$ $\cot \theta - \tan \theta = \left(\frac{-3}{-4} \right) - \left(\frac{-4}{-3} \right)$ $= -\frac{7}{12} \approx -0,58$	✓ SF A ✓ value of/waarde van y CA ✓ cot ratio/verhouding CA ✓ tan ratio/verhouding CA ✓ S CA (5)
3.3.1	Quadrant / Kwadrant III (3 rd / 3 ^{de}) and / en Quadrant / Kwadrant IV (4 th / 4 ^{de})	✓ quadrant/kwadrant III A ✓ quadrant/kwadrant IV A (2)
3.3.2	$\sin x = -\frac{1}{4}$ Ref. / Verwys. $\angle = 14,48^\circ$ $x = 180^\circ + 14,48^\circ$ or / of $x = 360^\circ - 14,48^\circ$ $\therefore x = 194,48^\circ$ or / of $x = 345,52^\circ$ OR / OF $\sin x = -\frac{1}{4}$ Ref. / Verwys. $\angle = 14,48^\circ$ $x = -14,48^\circ$ or / of $x \approx -180^\circ - (-14,48^\circ)$ $x \approx -165,52^\circ$	✓ ref. angle/verwys. hoek A ✓ value of/waarde van x CA ✓ value of/waarde van x CA OR / OF ✓ ref. angle/verwys. hoek A ✓ value of/waarde van x CA ✓ value of/waarde van x CA AO: full marks / volpunte (3)

QUESTION/VRAAG 4

4.1.1	$-\cos x$	✓ $-\cos x$	A (1)
4.1.2	1	✓ 1	A (1)
4.1.3	$\frac{\cos(\pi + x) \cdot \tan(360^\circ - x)}{[\cos^2(\pi + x) - \sin x \cdot \sin(360^\circ - x)]}$ $= \frac{(-\cos x)(-\tan x)}{\cos^2 x - \sin x(-\sin x)}$ $= \frac{(-\cos x)\left(-\frac{\sin x}{\cos x}\right)}{\cos^2 x + \sin^2 x}$ $= \frac{\sin x}{1} = \sin x$	✓ $-\tan x$ ✓ $\cos^2 x$ OR / OF $(-\cos x)^2$ ✓ $-\sin x$ ✓ I $\left(\frac{\sin x}{\cos x}\right)$ ✓ S	A A A A CA (5)
4.2.1	$\sin^2 x$	✓ $\sin^2 x$ OR / OF $\frac{1}{\frac{1}{\sin^2 x}}$	A (1)
4.2.2	$2 \sin x - 2 \sin x \cdot \cos^2 x = 2 \sin x(1 - \cos^2 x)$	✓ Factors / <i>Faktore</i>	A (1)
4.2.3	LHS/LK $\frac{2}{\operatorname{cosec}^2 x \cdot \sec x} \times \frac{1}{2 \sin x - 2 \sin x \cdot \cos^2 x}$ $= 2 \sin^2 x \cdot \cos x \times \frac{1}{2 \sin x(1 - \cos^2 x)}$ $= 2 \sin^2 x \cdot \cos x \times \frac{1}{2 \sin x \cdot \sin^2 x}$ $= \frac{\cos x}{\sin x}$ $= \cot x = \text{RHS/RK}$	✓ $\cos x$ OR / OF $\frac{1}{\cos x}$ ✓ I $\sin^2 x$ ✓ $\frac{\cos x}{\sin x}$ OR / OF $\frac{1}{\frac{\sin x}{\cos x}}$	A A A (3)

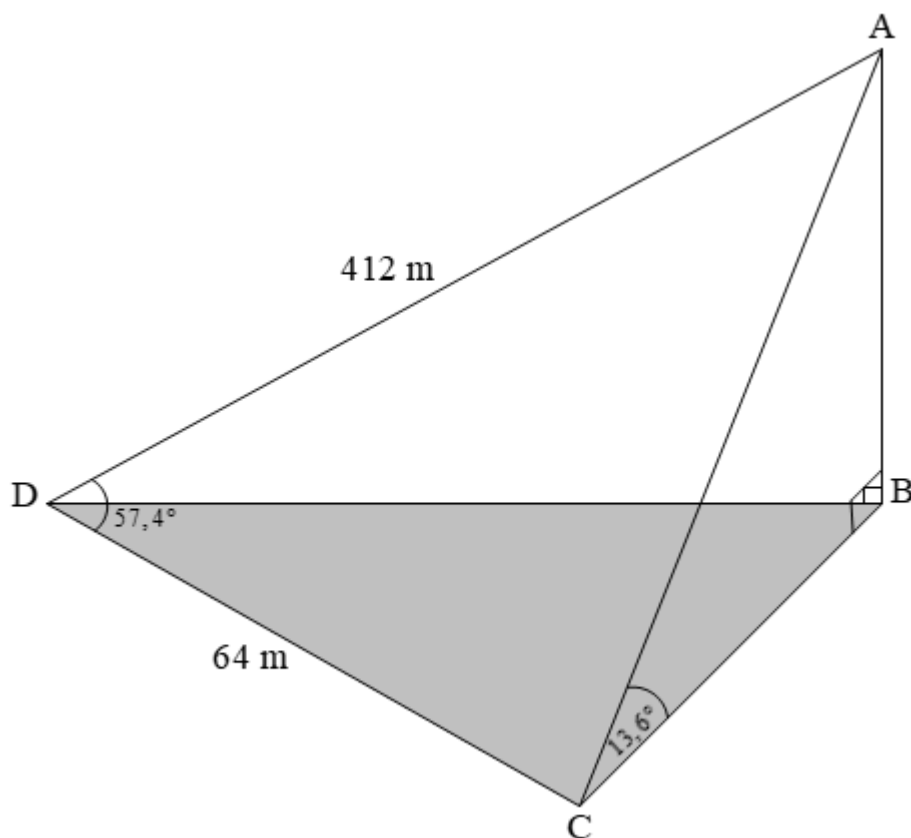
[12]

QUESTION/VRAAG 5

5.1	$a = 3$	✓ value of/ waarde van a	A (1)
5.2	$x = 90^\circ$	✓ equation/ vergelyking	A (1)
5.3	J (180°; -3)	✓ 180° ✓ -3	A A (2)
5.4	$y \in [-3; 3]$ OR / OF $-3 \leq y \leq 3$ OR / OF $y \geq -3$ and / en $y \leq 3$ OR / OF From -3 to 3 / Vanaf -3 tot 3	✓ end points/ eindpunte ✓ notation/ notasie OR / OF ✓ end points/ eindpunte ✓ notation/ notasie OR / OF ✓ end points/ eindpunte ✓ notation/ notasie OR / OF ✓ end points/ eindpunte ✓ notation/ notasie	A A A A A A A A (2)

5.5	$\tan m = 1,6$ $m \approx 57,99^\circ$ OR / OF $3 \cos m = 1,6$ $\cos m = \frac{1,6}{3} \quad \cos m = \frac{1,6}{3}$ $m \approx 57,77^\circ$ OR / OF $m = 180^\circ - 122^\circ$ $m = 58^\circ$ OR / OF $m = 90^\circ - 32^\circ$ $m = 58^\circ$ OR / OF $m = 45^\circ + 32^\circ$ $m = 58^\circ$	✓ substitution/vervanging A ✓ value of/ waarde van m CA OR / OF ✓ substitution/vervanging CA ✓ value of/ waarde van m CA OR / OF ✓ M A ✓ value of/ waarde van m CA OR / OF ✓ M A ✓ value of/ waarde van m CA OR / OF ✓ M A ✓ value of/ waarde van m CA AO: full marks / volpunte (2) NPR
5.6	$0^\circ \leq x \leq 58^\circ$ or / of $90^\circ < x \leq 122^\circ$ OR / OF $x \in [0^\circ ; 58^\circ] \cup (90^\circ ; 122^\circ]$ OR / OF x values from 0° to 58° or greater than 90° to 122° x waardes vanaf 0° tot 58° of groter as 90° tot 122°	✓ critical values/ kritiese waardes CA ✓ notation/ notasie A ✓ critical values/ kritiese waardes A ✓ notation/ notasie A OR / OF ✓ critical values/ kritiese waardes CA ✓ notation/ notasie A ✓ critical values/ kritiese waardes A ✓ notation/ notasie A OR / OF ✓ critical values/ kritiese waardes CA ✓ notation/ notasie A ✓ critical values/ kritiese waardes A ✓ notation/ notasie A (4)

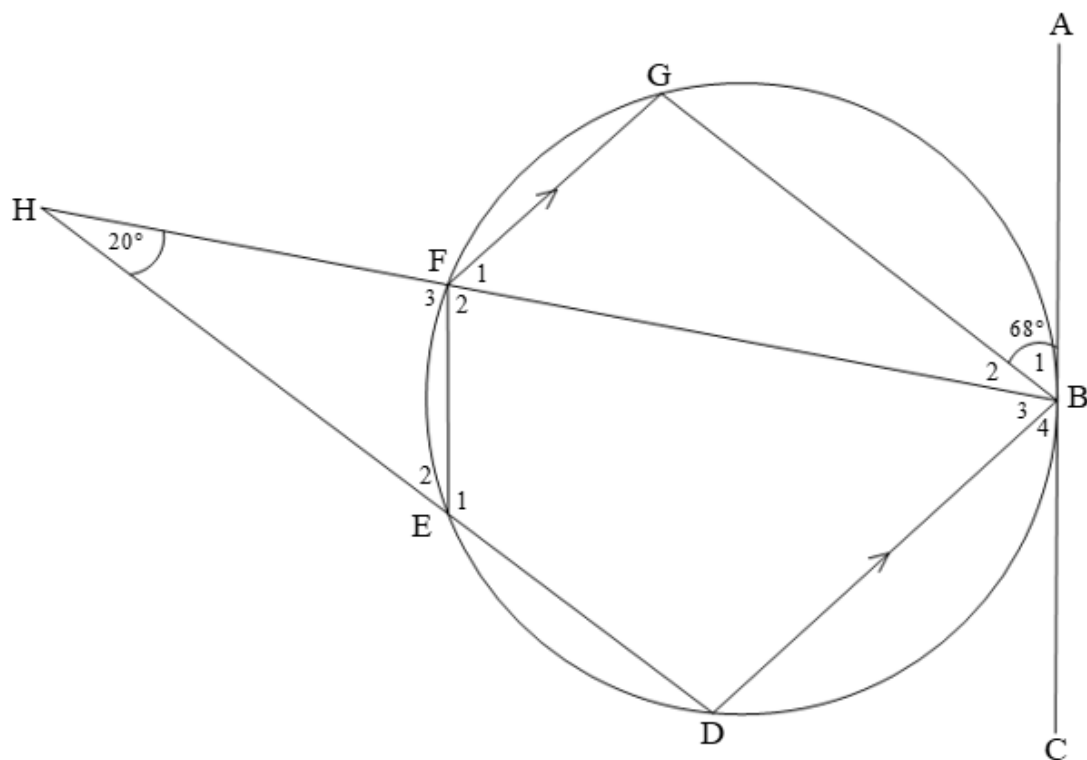
5.7	3	✓ 3	A (1)
5.8	$h(x) = -3 \cos x + 1$	✓ $-3 \cos x$ OR / OF $-a \cos x$ ✓ $+1$	CA A (2)
			[15]

QUESTION/VRAAG 6

6.1	$\text{Area of } \triangle ADC = \frac{1}{2} \cdot AD \cdot DC \cdot \sin D$ $= \frac{1}{2} (412)(64) \sin 57,4^\circ$ $\approx 11\,106,89 \text{ m}^2$	✓ area rule/ <i>Oppv. reël</i> A ✓ SF A ✓ value of area/ <i>waarde van oppv.</i> CA
6.2	$AC^2 = AD^2 + DC^2 - 2 AD \cdot DC \cdot \cos \hat{D}$ $AC^2 = (412)^2 + (64)^2 - 2(412)(64) \cdot \cos 57,4^\circ$ $AC^2 = 145\,427,3839$ $AC \approx 381,35 \text{ m}^2$ $\sin 13,6^\circ = \frac{AB}{381,35} \quad \text{OR / OF} \quad \frac{AB}{\sin 16,6^\circ} = \frac{381,35}{\sin 90^\circ}$ $AB = 381,35 \cdot \sin 13,6^\circ$ $AB \approx 89,67 \text{ m}$	✓ cos rule/ <i>reël</i> A ✓ SF A ✓ S CA ✓ length of/ <i>lengte van</i> AC CA ✓ sin ratio or rule/ <i>verhouding of reël</i> CA ✓ height/ <i>hoogte</i> AB CA

[9]

QUESTION/VRAAG 7



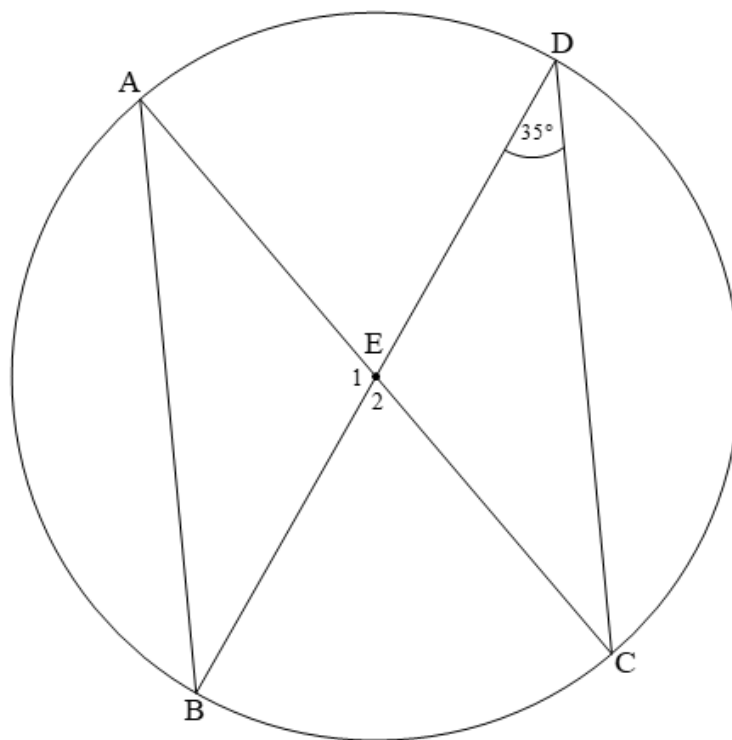
7.1	opposite / <i>teenoorstaande</i>	✓ ST	A (1)
7.2.1	$\hat{F}_1 = 68^\circ$ $\left(\begin{array}{l} \text{Tan – chord Theorem /} \\ \text{Raakl. – Koordstelling} \end{array} \right)$	✓ ST ✓ RE	A A (2)
7.2.2	$\hat{B}_3 = 68^\circ$ (Alt. $\angle$ s / <i>Verw. $\angle$e</i> =; FG $\parallel$ BD)	✓ ST ✓ RE	CA A (2)
7.2.3	$\hat{E}_2 = 68^\circ$ $\left(\begin{array}{l} \text{Ext. } \angle \text{ cyclic. quad /} \\ \text{Buite } \angle \text{ kdvh.} \end{array} \right)$ OR / OF $\hat{E}_1 = 112^\circ$ $\left(\begin{array}{l} \text{Opp. } \angle \text{s of cyclic quad. /} \\ \text{Teenoorst. } \angle \text{e kdvh.} \end{array} \right)$ $\hat{E}_2 = 68^\circ$ $\left(\begin{array}{l} \angle \text{s on str. line /} \\ \angle \text{e op reguitlyn} \end{array} \right)$	✓ ST ✓ RE OR / OF ✓ ST/RE ✓ ST	CA A CA CA (2)

7.2.4	$\hat{F}_2 = 20^\circ + 68^\circ \quad (\text{Ext. } \angle \Delta / \text{Buite } \angle \Delta)$ $\therefore \hat{F}_2 = 88^\circ$ OR / OF $\hat{F}_3 = 92^\circ \quad (\text{Int. } \angle \text{s } \Delta / \text{Binne } \angle \text{e } \Delta)$ $\hat{F}_2 = 88^\circ \quad \left(\begin{array}{l} \angle \text{s on straight line /} \\ \angle \text{e op reguitlyn} \end{array} \right)$ OR / OF $\hat{D} = 180^\circ - 20^\circ - 68^\circ \quad (\text{Int. } \angle \text{s } \Delta / \text{Binne } \angle \text{e } \Delta)$ $\hat{D} = 92^\circ$ $\therefore \hat{F}_2 + 92^\circ = 180^\circ \quad \left(\begin{array}{l} \text{Opp. } \angle \text{s cyclic quad /} \\ \text{Teenoorst. } \angle \text{e kdvh.} \end{array} \right)$ $\therefore \hat{F}_2 = 88^\circ$	✓ RE A ✓ ST CA OR / OF ✓ RE (Any of the two / enige van die twee) A ✓ ST CA OR / OF ✓ RE (Any of the two / enige van die twee) A ✓ ST CA (2)
7.2.5 #	$\hat{D} = 180^\circ - 88^\circ \quad \left(\begin{array}{l} \text{Opp. } \angle \text{s cyclic. quad. /} \\ \text{Teenoorts. } \angle \text{e kdvh.} \end{array} \right)$ $\hat{D} = 92^\circ$ OR / OF $\hat{D} = 92^\circ \quad \left(\begin{array}{l} \text{Int } \angle \text{s } \Delta / \\ \text{Binne. } \angle \text{e } \Delta \end{array} \right)$ OR / OF $\hat{D} = \hat{F}_3 = 92^\circ \quad \left(\begin{array}{l} \text{Ext. } \angle \text{ cyclic. quad. /} \\ \text{Buite } \angle \text{ kdvh.} \end{array} \right)$	✓ RE A ✓ ST CA OR / OF ✓ ST CA ✓ RE A OR / OF ✓ ST CA ✓ RE A (2)

[11]

8.1.3	Ext. $\angle$ = opp. int. $\angle$ / Buite $\angle$ = teenoorst. binne $\angle$	✓ RE	A
	OR / OF	OR / OF	
	Conv. ext. $\angle$ cyclic quad / Omg. buite $\angle$ kdvh.	✓ RE	A
	OR / OF	OR / OF	
	Opp. $\angle$ s suppl. / Teenoorst. $\angle$ e suppl.	✓ RE	A
			(1)

8.2

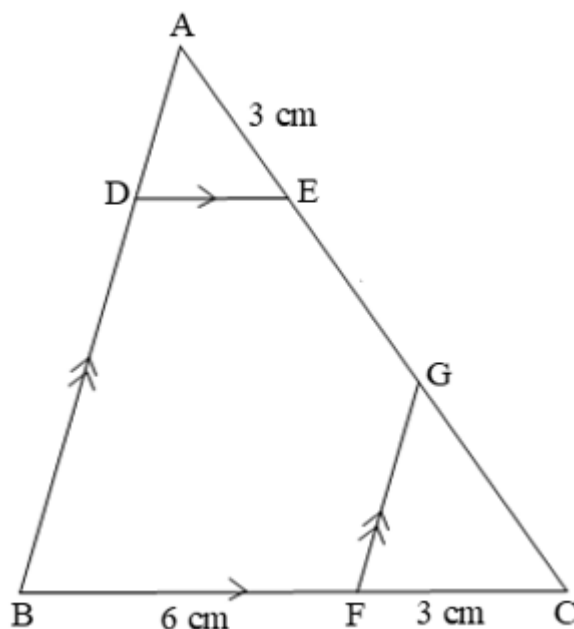


8.2.1	$\hat{A} = 35^\circ$ ($\angle$ s opp. = sides / $\angle$ teen. = sye) OR / OF	✓ ST	A
	($\angle$ s in same segm. / $\angle$ e in dies. segm.)	✓ RE	A
	$\hat{B} = 35^\circ$ ($\angle$ s opp. = sides / $\angle$ teen. = sye) OR / OF	✓ ST	A
	($\angle$ s in same segm. / $\angle$ e in dies. segm.)	✓ RE	A
	$\hat{C} = 35^\circ$ ($\angle$ s opp. = sides / $\angle$ teen. = sye) OR / OF	✓ ST	A
	($\angle$ s in same segm. / $\angle$ e in dies. segm.)	✓ RE	A
	Note: The reason mark(s) are dependent on the order in which the angles are stated. / Nota: Die punt(e) vir die rede is afhanklik van die volgorde waarin die hoeke bepaal is.		(6)

8.2.2	$\hat{E}_2 = 70^\circ$ (Ext. /Buite $\angle \Delta$)	✓ ST	CA
	OR / OF	✓ RE	A
	$\hat{E}_2 = 70^\circ$ $\left(\begin{array}{l} \angle \text{ at centre} = 2 \times \angle \text{ at circumf. / } \\ \text{midpts } \angle = 2 \times \text{omtreks } \angle \end{array} \right)$	✓ ST	CA
	OR / OF	✓ RE	A
	$\hat{E}_1 = 110^\circ$ (Int. $\angle$ s / Binne $\angle$ e Δ)	✓ RE (Any of the two / enige van die twee)	A
	$\hat{E}_2 = 70^\circ$ ($\angle$ s on str. line / $\angle$ e op reguitlyn)	✓ ST	CA
			(2)
8.2.3	S $\angle$ S OR / OF $\angle \angle$ S	✓ RE	A
			(1)

[16]

QUESTION/VRAAG 9

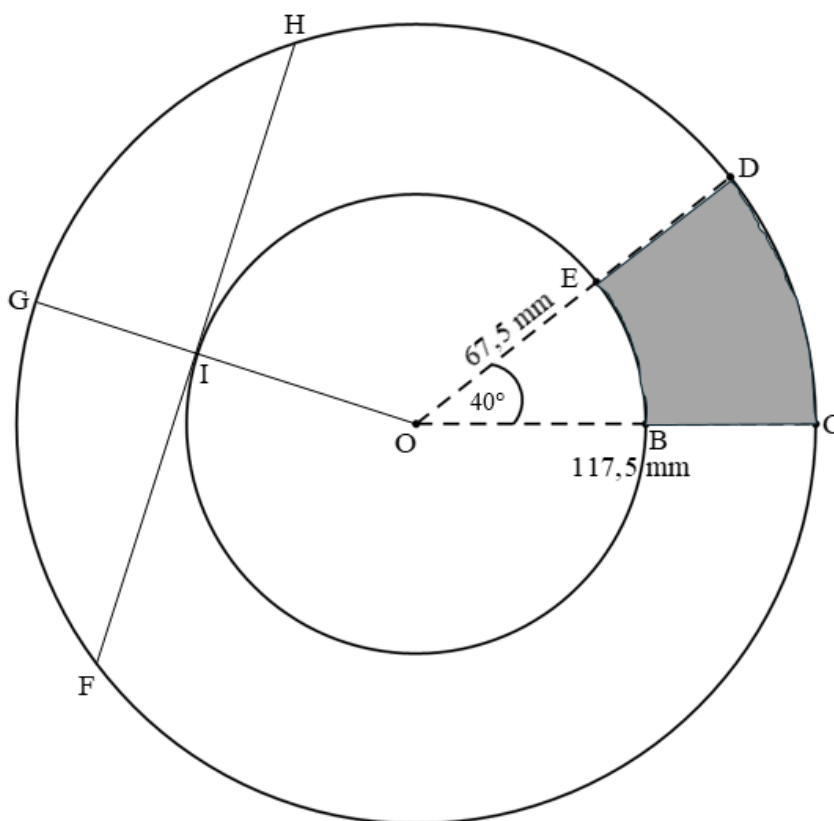


9.1	$\frac{EC}{AE} = \frac{DB}{AD} \quad \left(\begin{array}{l} \text{Prop. Th, } DE \parallel BC / \\ \text{Eweredigheid st, } DE \parallel BC \end{array} \right)$ OR / OF $\frac{EC}{AE} = \frac{DB}{AD} \quad \left(\begin{array}{l} \text{Line parallel to one side } \Delta / \\ \text{Lyn ewewydig aan een sy } \Delta \end{array} \right)$ $\frac{EC}{3} = \frac{3}{1}$ $EC = 9 \text{ cm}$	✓ ST A ✓ RE A OR / OF ✓ ST A ✓ RE A ✓ SF A ✓ Length / Lengte EC CA AO: $\frac{1}{4}$ (4)
9.2	$\frac{CF}{CB} = \frac{CG}{CA} \quad \left(\begin{array}{l} \text{Prop. Th, } GF \parallel AB / \\ \text{Eweredigheid st, } GF \parallel AB \end{array} \right)$ OR / OF $\frac{CF}{CB} = \frac{CG}{CA} \quad \left(\begin{array}{l} \text{Line parallel to one side } \Delta / \\ \text{Lyn ewewydig aan een sy } \Delta \end{array} \right)$ $\frac{3}{9} = \frac{CG}{12}$ $\therefore CG = 4 \text{ cm}$	✓ ST A OR / OF ✓ ST A ✓ Substitution/vervanging CA ✓ length / lengte GC CA AO: $\frac{1}{3}$ (3)

9.3	In $\triangle ADE$ & $\triangle ABC$:		
	(1) $\hat{A} = \hat{A}$ (common / gemene $\angle$)	✓ ST	A
	(2) $\hat{ADE} = \hat{ABC}$ (corr. $\angle$ s/ooreenk. $\angle$ e DE $\parallel$ BC)	✓ ST	A
	(3) $\hat{AED} = \hat{ACB}$ ($\angle$ s of / $\angle$ e van $\triangle$) OR / OF (corr. $\angle$ s / ooreenk. $\angle$ e; DE $\parallel$ BC)	✓ ST	A
	$\therefore \triangle ADE \parallel \triangle ABC$ ($\angle \angle \angle$)		(3)
			[10]

QUESTION/VRAAG 10

10.1



10.1.1	$\theta = 40^\circ \times \frac{\pi}{180^\circ}$ $= \frac{2}{9} \pi \approx 0,70 \text{ rad}$	✓ answer / antwoord A (1)
10.1.2	$s = r\theta$ $s = (117,5) \left(\frac{2}{9} \pi \right) \quad \text{OR / OF} = (117,5)(0,70)$ $s = \frac{235}{9} \pi \approx 82,03 \text{ mm} \quad \approx 82,25 \text{ mm}$ OR / OF $s = \frac{40^\circ}{360^\circ} \times 2\pi r$ $s = \frac{40^\circ}{360^\circ} \times 2\pi(117,5)$ $s \approx 82,03 \text{ mm}$	✓ F A ✓ SF for / vir θ CA ✓ answer / antwoord CA OR / OF ✓ F A ✓ SF for / vir θ A ✓ answer / antwoord CA (3)

10.1.3	$\text{Area/Oppv.} = \frac{r^2 \theta}{2}$	✓ F A
	$\text{Area/Oppv.} = \frac{(67,5)^2 \left(\frac{2}{9} \pi\right)}{2} \quad \text{OR / OF} = \frac{(67,5)^2 (0,70)}{2}$	✓ SF for / vir θ CA
	$\text{Area/Oppv.} = \frac{2025}{4} \pi$	
	$\text{Area/Oppv.} \approx 1590,43 \text{ mm}^2 \quad \approx 1594,69 \text{ mm}^2$ OR / OF	✓ Area/Oppv. CA OR / OF
	$s = r\theta$	
	$s = (67,5) \left(\frac{2}{9} \pi\right) \quad \text{OR / OF} = (67,5)(0,70)$	
	$s = 15 \pi \approx 47,12 \text{ mm} \quad \approx 47,25 \text{ mm}$	
	$\text{Area/Oppv.} = \frac{rs}{2}$	✓ F A
	$\text{Area/Oppv.} = \frac{(67,5)(15 \pi)}{2} \quad \text{OR / OF} = \frac{(67,5)(47,25)}{2}$	✓ SF for / vir s CA
	$\text{Area/Oppv.} = \frac{2025}{4} \pi \approx 1590,43 \text{ mm}^2 \quad \approx 1594,69 \text{ mm}^2$ OR / OF	✓ Area/Oppv. CA OR / OF
	$\text{Area/Oppv.} = \frac{40^\circ}{360^\circ} \pi r^2$	✓ F A
	$\text{Area/Oppv.} = \frac{40^\circ}{360^\circ} \pi (67,5)^2$	✓ SF A
	$\text{Area/Oppv.} = \frac{2025}{4} \pi \approx 1590,43 \text{ mm}^2$	✓ Area/Oppv. CA
		(3)

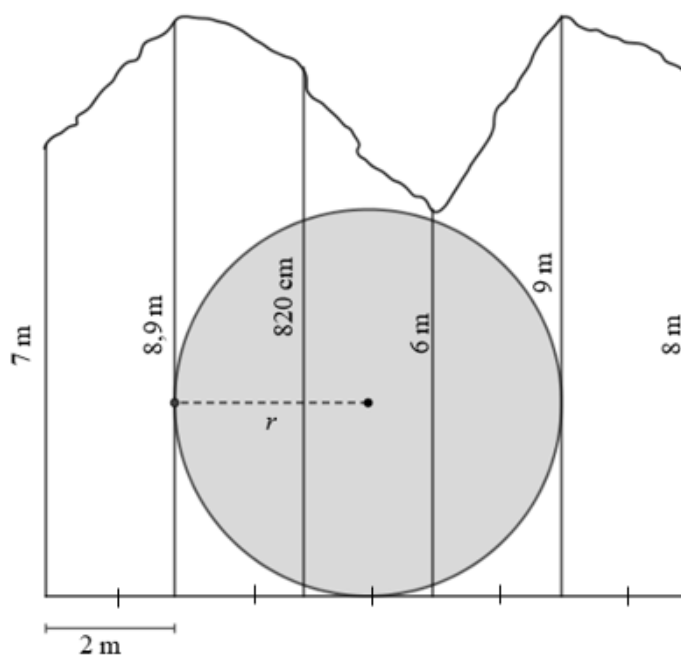
10.1.4	$GI = h = 117,5 - 67, = 50 \text{ mm}$ $d = 2 \times 117,5 = 235 \text{ mm}$ $4h^2 - 4dh + x^2 = 0$ $4(50)^2 - 4(235)(50) + x^2 = 0$ $x^2 = 37000$ $x = 10\sqrt{370} \approx 192,35 \text{ mm}$ OR / OF $HI^2 = HO^2 - OI^2$ $HI^2 = (117,5)^2 - (67,5)^2$ $HI^2 = 9250$ $HI = 5\sqrt{370} \approx 96,18$ $x = 10\sqrt{370} \approx 192,35 \text{ mm}$ OR / OF $d = 235$ $h = 235 - (117,5 - 67,5) = 185$ $4h^2 - 4dh + x^2 = 0$ $4(185)^2 - 4(235)(185) + x^2 = 0$ $x^2 = 37000$ $x = 10\sqrt{370} \approx 192,35 \text{ mm}$	values of / ✓ waardes van h & d A ✓ F A ✓ SF CA ✓ Value of / waarde van x CA OR / OF ✓ M A ✓ SF A ✓ Length / Lengte HI CA ✓ Value of / waarde van x CA OR / OF values of / ✓ waardes van h & d A ✓ F A ✓ SF CA ✓ Value of / waarde van x CA (4)
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10.1.5	Area sector OCD / Oppv. sektor OCD = $\frac{r^2\theta}{2}$ # $= \frac{(117,5)^2 \left(\frac{2}{9}\pi\right)}{2}$ OR / OF $= \frac{(117,5)^2 (0,70)}{2}$ $= \frac{55225}{36}\pi$ $\approx 4819,29 \text{ mm}^2$ $\approx 4832,19 \text{ mm}^2$ $A_{\text{BCDE}} = \frac{55225}{36}\pi - \frac{2025}{4}\pi$ $A_{\text{BCDE}} = 3228,86 \text{ mm}^2$ Area of outer circle / Oppv. van buitesirkel Area of BCDE / Oppv. van BCDE $\frac{\frac{55225}{4}\pi}{\frac{9250}{4}\pi}$ $\approx 13,43$ OR / OF Area/Oppv. = $\frac{rs}{2}$ $\frac{(117,5)\left(\frac{235}{9}\pi\right)}{2}$ Area/Oppv. $\approx 4819,29 \text{ mm}^2$ $A = \frac{55225}{36}\pi - \frac{2025}{4}\pi$ $A = 3228,86 \text{ mm}^2$ Area of outer circle / Oppv. van buitesirkel Area of BCDE / Oppv. van BCDE $\frac{\frac{55225}{4}\pi}{\frac{9250}{4}\pi}$ $\approx 13,43$ OR / OF	✓ SF CA ✓ Area/Oppervlakte CA ✓ M A ✓ Area of / Oppv. van BCDE CA ✓ Value of ratio / waarde van verhouding OR / OF CA ✓ SF CA ✓ Area / Oppv. CA ✓ M A ✓ Area / Oppv. CA ✓ Value of ratio / waarde van verhouding OR / OF CA
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	$A = \pi r^2$ $A = \pi (67,50)^2$ $A = \frac{18225}{4} \pi \approx 14313,88 \text{ mm}^2$ $\text{Area/Oppv.} = \frac{40^\circ}{360^\circ} \left[\frac{55225}{4} \pi - \frac{18225}{4} \pi \right]$ $\text{Area/Oppv.} = \frac{9250}{9} \pi \approx 3228,86 \text{ mm}^2$ $\frac{\text{Area of outer circle}}{\text{Area of BCDE}} / \frac{\text{Oppv. van buitesirkel}}{\text{Oppv. van BCDE}}$ $= \frac{\frac{55225}{4} \pi}{\frac{9250}{4} \pi}$ $\approx 13,43$	✓ SF A ✓ Area / Oppv. CA ✓ M A ✓ Area / Oppv. CA Value of ratio / ✓ waarde van verhouding CA (5)
10.2.1	$\omega = 2 \pi n$ $90 \pi = 2 \pi n$ $n = \frac{90 \pi}{2 \pi}$ $n = 45 \text{ rev / s}$	✓ F A ✓ SF A ✓ Rota. freq. / Rota. Frekw. CA (3)
10.2.2	$v = \pi D n$ $v = \pi (235)(45)$ $v = 10575 \pi \approx 33222,34 \text{ mm / s}$ OR / OF $v = \omega r$ $v = (90 \pi)(117,5) \text{ OR / OF } v = (282,74)(117,5)$ $v = 10575 \pi$ $v \approx 33222,34 \text{ mm / s OR / OF } v \approx 33221,95 \text{ mm / s}$	✓ F A ✓ SF CA ✓ Circ. velocity / Omtr. snelheid CA OR / OF ✓ F A ✓ SF A ✓ Circ. velocity / Omtr. snelheid CA (3)

QUESTION / VRAAG 11

11.1

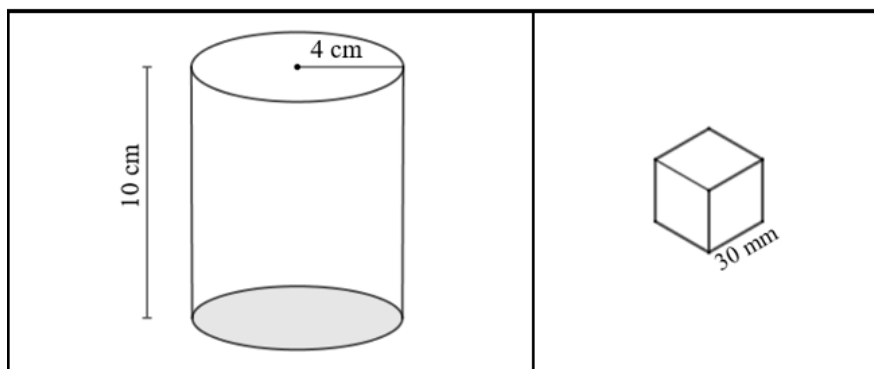


11.1.1	10 m	✓ 10 m	A (1)
11.1.2	$r = 3 \text{ m}$	✓ 3 m	A (1)
11.1.3	$820 \text{ cm} \div 100 = 8,2 \text{ m}$	✓ 8,2 m	A (1)

11.1.4	Area of irregular fig./Oppv. van onreëlmatige fig.		
#	$A_T = a \left(\frac{o_1 + o_n}{2} + o_2 + o_3 + \dots + o_{n-1} \right)$ $= (2) \left(\frac{7+8}{2} + 8,9 + 8,2 + 6 + 9 \right)$ $= \frac{396}{5} = 79,2 \text{ m}^2$	✓ SF	CA
	Area of circle/Oppv. van sirkel		
	$= \pi (3)^2 = 9\pi \approx 28,27 \text{ m}^2$	✓ area of circle/ oppv. van sirkel	CA
	Area to be paved/Oppv. wat gepavlei moet word		
	$= 79,2 - 9\pi = 50,93 \text{ m}^2$	✓ area to be paved / oppv. om te plavei	CA
	No. of paving bricks needed/Getal plaveistene benodig		
	Area covered by bricks / Oppv. bedek deur stene		
	$= \frac{50,93}{0,02} \approx 2547 \quad \text{OR / OF} \quad = 0,02 \times 2500 = 50 \text{ m}^2$	✓ bricks needed / bakstene benodig OR / OF Area of bricks / Oppv. van stene	CA
	$\therefore$ 2 500 paving bricks will NOT be sufficient. /plavleistene sal nie genoeg wees nie. OR / OF	✓ conclusion/afleiding OR / OF	CA

Area of irregular fig./Oppv. van <i>onreëlmatige fig.</i>: $A_T = a(m_1 + m_2 + m_3 + \dots + m_n)$ $= (2) \left(\frac{7+8,9}{2} + \frac{8,9+8,2}{2} + \frac{8,2+6}{2} + \frac{6+9}{2} + \frac{9+8}{2} \right)$ $= \frac{396}{5} = 79\frac{1}{5} = 79,2 \text{ m}^2$ Area of circle/ Oppv. van <i>sirkel</i> $= \pi(3)^2 = 9\pi \approx 28,27 \text{ m}^2$ Area to be paved/ Oppv. wat gepavlei moet word $= 79,2 - 9\pi = 50,93 \text{ m}^2$ No. of paving bricks needed/Getal plaveistene benodig $= \frac{50,93}{0,02} \approx 2547$ Area covered by bricks / Oppv. bedek deur stene $= 0,02 \times 2500 = 50 \text{ m}^2$ $\therefore$ 2 500 paving bricks will NOT be sufficient/ plaveistene sal nie genoeg wees nie	✓ SF CA ✓ area of irregular figure/oppv. van <i>onreëlmatige fig.</i> CA ✓ area of circle/ oppv. van <i>sirkel</i> CA ✓ area to be paved / oppv. om te plavei CA ✓ bricks needed / bakstene benodig OR / OF CA Area of bricks / Oppv. van stene ✓ conclusion/afleiding CA (6)
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11.2



11.2.1	$30\text{ mm} \div 10 = 3\text{ cm}$	✓ 3 cm	A (1)
11.2.2	Vol. of cylinder/<i>van silinder</i> $= \pi(4)^2(10)$ $= 160\pi\text{ cm}^3 \approx 502,65\text{ cm}^3$ Vol. of 1 ice cube / <i>van 1 ysblokkie</i> $= (3)^3$ $= 27\text{ cm}^3$ Vol. of 3 ice cubes/<i>van 3 ysblokkies</i> $= 3 \times 27$ $= 81\text{ cm}^3$ Vol. of remaining space / <i>van oorblywende spasie</i> $= 160\pi - (81 + 250)$ $\approx 171,65\text{ cm}^3$ $\therefore 171,65\text{ m}\ell$ OR / OF	✓ SF ✓ Vol. of cylinder / <i>van silinder</i> ✓ Vol. of 1 cube / <i>van 1 kubus</i> ✓ Vol. of 3 cubes / <i>van 3 kubusse</i> ✓ M ✓ Vol. remaining / <i>wat oorbly</i> OR / OF	A CA CA CA A CA

Vol. of cylinder/<i>van silinder</i> $= \pi (40)^2 (100)$ $= 160\,000 \pi \text{ mm}^3 \approx 502\,654,82 \text{ mm}^3$ Vol. of 1 ice cube / <i>van 1 ysblokkie</i> $= (30)^3$ $= 27\,000 \text{ mm}^3$ Vol. of 3 ice cubes/<i>van 3 ysblokkies</i> $= 3 \times 27\,000$ $= 81\,000 \text{ mm}^3$ Vol. of remaining space / <i>van oorblywende spasie</i> $= 160\,000 \pi - (81\,000 + 250\,000)$ $\approx 171\,654,82 \text{ mm}^3$ $\therefore 171,65 \text{ m}\ell$	✓ SF A ✓ Vol. of cylinder / <i>van silinder</i> CA ✓ Vol. of 1 cube / <i>van 1 kubus</i> CA ✓ Vol. of 3 cubes / <i>van 3 kubusse</i> CA ✓ M A ✓ Vol. remaining / <i>wat oorbly</i> CA (6)
[16]	

TOTAL/TOTAAL:150
